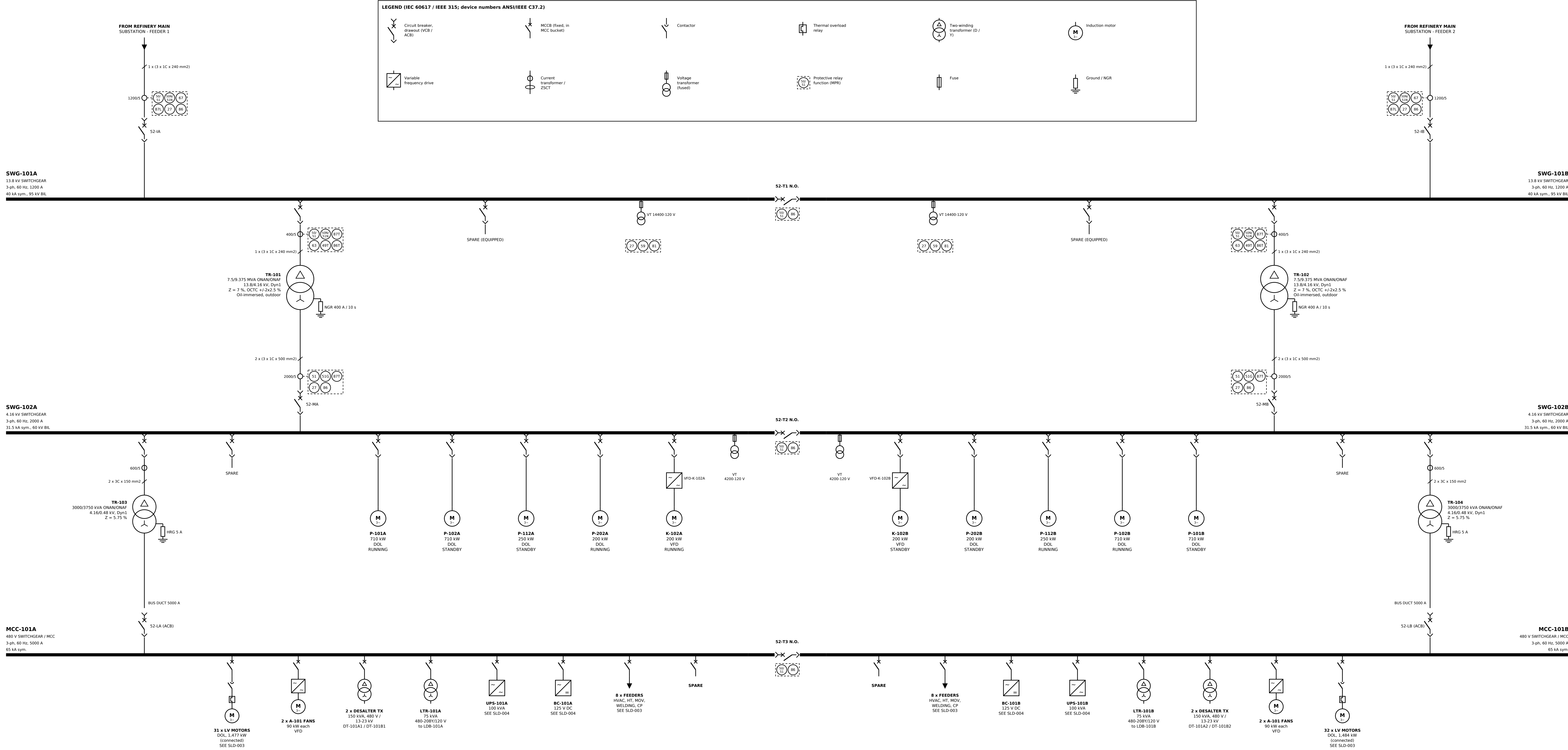




MAXIMUM DEMAND SUMMARY (MD = C + 0.3 I + 0.1 S)							
BUS	C kW	I kW	S kW	MD kW	MD kvar	MD kVA	PF
MCC-101A	1,082	127	559	1,176	641	1,340	0.88
MCC-101B	1,079	124	575	1,173	632	1,333	0.88
SWG-102A (MV motors)	985	0	836	1,068	524	1,190	0.90
SWG-102A (incl. LV)				2,257	1,190	2,551	0.88
SWG-102B (MV motors)	836	0	985	934	487	1,054	0.89
SWG-102B (incl. LV)				2,120	1,145	2,409	0.88
UNIT TOTAL at 13.8 kV				4,411	2,475	5,058	0.87

MAIN EQUIPMENT RATINGS			
TAG	DESCRIPTION	RATING	SC / LOAD
SWG-101A/B	13.8 kV metal-clad, VCB	1200 A	40 kA
TR-101/102	7.5/9.375 MVA 13.8/4.16 kV	Z 7 %	66 % MD
SWG-102A/B	4.16 kV metal-clad, VCB, AR	2000 A	31.5 kA
TR-103/104	3000/3750 kVA 4.16/0.48 kV	Z 5.75 %	89 % MD
MCC-101A/B	480 V LV swgr + MCC	5000 A	65 kA
UPS-101A/B	2 x 100 kVA, 30 min	VRLA	350 Ah

FAULT LEVELS (max. case)			
BUS	I _k ⁺ kA	I _p kA	RATED kA
13.8 kV	32.2	82.9	40
4.16 kV	15.8	40.3	31.5
0.48 kV	52.2	122.7	65

MOTOR STARTING (largest DOL motor):
P-101A, 710 kW DOL at 4.16 kV:
bus dip 5.3 % (limit 10 %),
terminal dip 5.8 % (limit 15 %)

- NOTES**
- Normal operation: both incomers closed, bus ties NORMALLY OPEN (N.O.) at 13.8 kV, 4.16 kV and 480 V.
 - Each transformer sized for the total unit maximum demand (one out of service, tie closed): ONAN => MD, ONAF => 1.25 x MD (25 % future). See CFU-000-EL-CAL-001.
 - Tie breakers: 2-out-of-3 electrical/key interlock per board (no paralleling of transformers); auto-transfer (open transition) on loss of incomer, with residual-voltage supervision (27R).
 - 4.16 kV system low-resistance grounded (400 A); 480 V high-resistance grounded (5 A, alarm).
 - Pump A/B (2 x 100 %) are always fed from opposite bus sections; the normally running unit of each pair is selected to balance the buses (load list CFU-000-EL-LDL-001).
 - No emergency diesel generator: essential loads (DCS/SIS/F&G/telecom) on 2 x 100 % UPS, 30 min autonomy; unit fails safe on total power loss. See SLD-004 note 1.
 - Detailed MV / LV / UPS SLDs: CFU-000-EL-SLD-002 / -003 (2 sheets) / -004.
 - Short-circuit ratings: symmetrical interrupting kA; utility 31.5 kA at 13.8 kV (basis).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
OVERALL KEY SINGLE LINE DIAGRAM 13.8 / 4.16 / 0.48 KV DISTRIBUTION - SUBSTATION SS-100			
DISCIPLINE	STAGE	SCALE	SHEET
ELECTRICAL	FEED	NTS	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-000-EL-SLD-001			A